



IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

AI Deployment and automation

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION

June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Aman Sharma

Enrollment / Roll No.
2410030051

Program & Section
B.Tech (CSE) | 3CSE1

HOST ORGANIZATION
EduSkills Academy
Supported by AICTE National Internship Portal

Candidate & Program Overview

Candidate Details

Student Name: AMAN SHARMA

Roll Number: 2410030051

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026-2027 | Section: 3CSE1

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: AICTE – EduSkills Virtual Internship

Specialization Domain: AI Deployment & Automation

Duration: 8 Weeks (June 2026 – August 2026)

Certificate ID: 2026-E31C2B5F43

Offer Letter Ref: C17Y2026M081522484

Organization Profile: EduSkills Academy

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization operating under the vision of "*Nation Building Through Skills*".



AICTE Collaboration: Operates under the patronage of the AICTE National Internship Portal to empower engineering talent across India.



Industry Alignment: Partners with leading global tech organizations to bridge academia and industry gaps in emerging tech.



Core Focus Domains: Artificial Intelligence, MLOps, Cloud Computing, Cybersecurity, and Enterprise Automation.

Problem Statement & Industry Context

The Industry Challenge

While academic curriculum extensively covers building Machine Learning and Deep Learning models, students often lack hands-on exposure to deploying models into real-world production systems.

A high-performing ML model in a Jupiter Notebook provides zero business value until it is exposed as a reliable, scalable, monitorable, and secure API service.

The Solution Scope

This internship addresses the complete end-to-end lifecycle of production AI: model serving, container packaging, cloud hosting, automated scaling, observability, and retraining triggers.

Mastering industry-grade tools (FastAPI, Docker, Kubernetes, LangChain, SageMaker) to transform research prototypes into enterprise AI solutions.

Core Project Objectives



API & Containerization

Learn advanced model serving with FastAPI (async operations) and build portable, production-ready Docker containers for MLOps.



Cloud & Orchestration

Deploy AI/GenAI models on Amazon SageMaker, Vertex AI, and Azure ML, and manage multi-container pods using Kubernetes.



Agentic Workflows & RAG

Construct intelligent AI agents and business automation pipelines using LangChain, LlamaIndex, LangGraph, and Vector Databases.



Observability & Drift

Implement continuous monitoring and logging using Prometheus, Grafana, and Evidently AI for automated drift detection.

Technologies & MLOps Toolchain



Serving & Apps

Languages & APIs: Python, FastAPI
(Async REST Services)

Front end Integration: Streamlit,
Gradio Web UI



DevOps & Cloud

Containers: Docker, Kubernetes
Cluster (Pods, Deployments)

Cloud Platforms: AWS SageMaker,
Google Vertex AI, Azure ML



GenAI & Agents

Frameworks: LangChain,
LlamaIndex, LangGraph

Serving & RAG: vLLM, Vector DBs,
Quantization



Monitoring & Drift

Observability: Prometheus, Grafana
Dashboards

Drift Analysis: Evidently AI for data
drift & retraining

Production AI Architecture



Model Layer: Base ML / LLM models providing intelligence.



API Layer: FastAPI wrapper for sync & async inference calls.



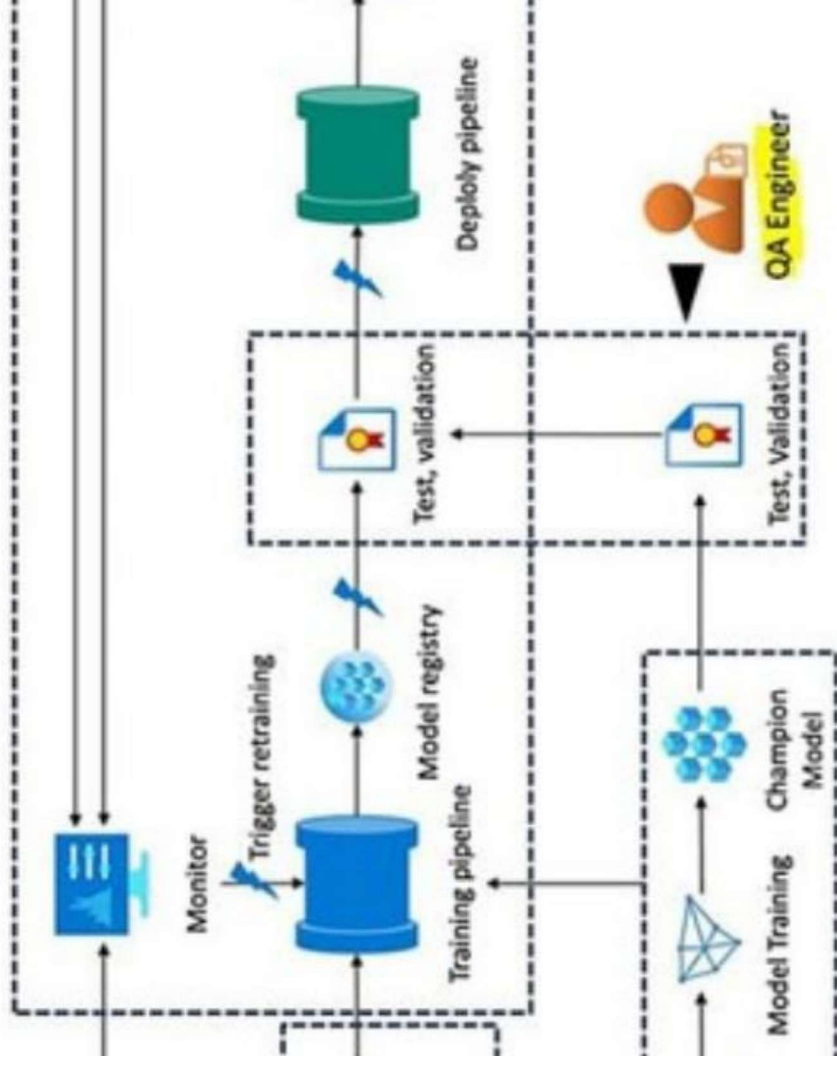
Container & Cloud: Docker packaged containers orchestrated via Kubernetes on AWS/Azure/Vertex.



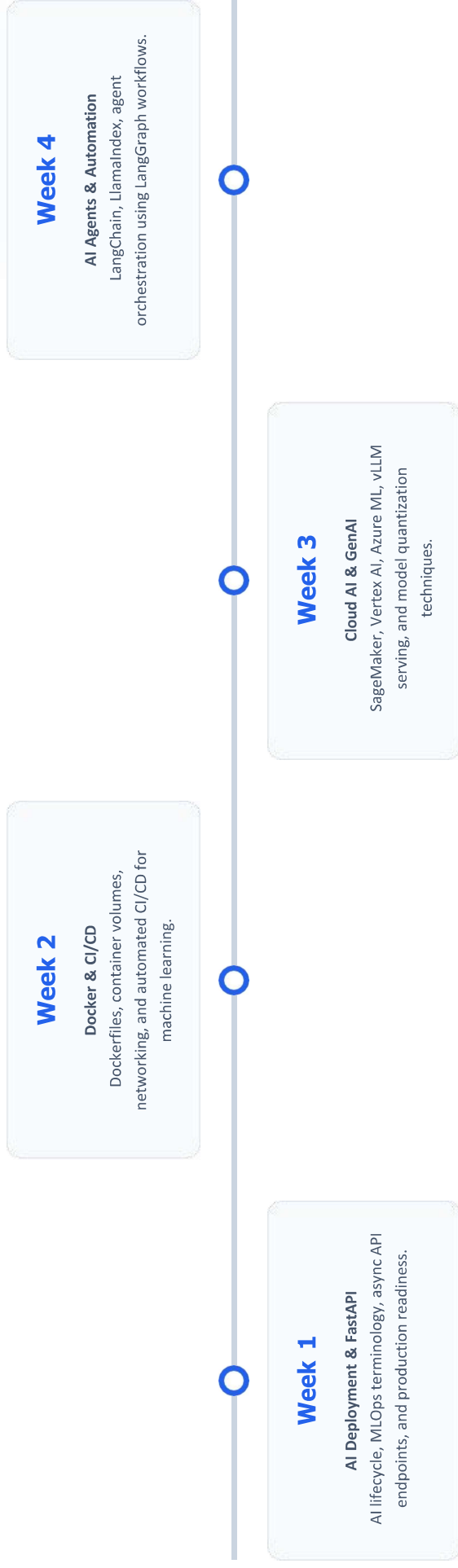
Agent & App Layer: LangChain & LlamaIndex workflows integrated with Streamlit interfaces.



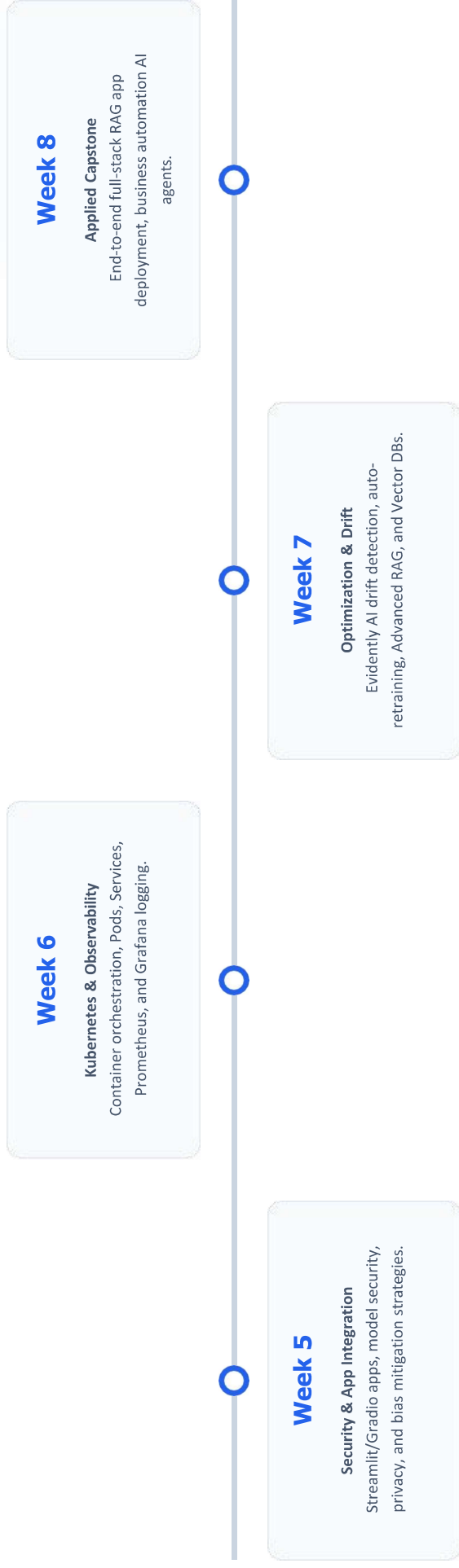
Observability Layer: Real-time tracking with Prometheus, Grafana, and Evidently AI.



Curriculum Timeline (Weeks 1 to 4)



Curriculum Timeline (Weeks 5 to 8)



Key Projects & Demonstrated Skills

- ✓ **Production-Ready API Development:** Developed high-throughput RESTful endpoints using FastAPI with async support for AI inference.
- ✓ **Containerization & Orchestration:** Package ML models into lightweight Docker containers and orchestrating them via Kubernetes Pods & Services.
- ✓ **Full-Stack RAG Application:** Built an end-to-end Retrieval-Augmented Generation application utilizing Vector Databases, FastAPI backend, and Streamlit frontend.
- ✓ **Agentic Workflows for Automation:** Designed multi-step AI agents using LangChain and LangGraph for real-world enterprise business automation.
- ✓ **Model Governance & Drift Monitoring:** Configured Evidently AI alongside Prometheus and Grafana dashboards to track prediction drift and trigger retraining.

Certification & Official Verification

Verification Metric		Official Record / Credential Details	
Internship Title	AI Deployment & Automation Virtual Internship		
Issuing Organization	EduSkills Academy (ISO 9001:2015 & ISO 20000-1:2018 Certified)		
Academic Patronage	AICTE National Internship Portal & EduSkills Foundation		
Certificate ID	2026-E31C2B5F43 (Issued: Jul 31, 2026)		
Offer Letter Ref	C17Y2026M081522484		
AICTE Student ID	STU69e50fbaba77a1776619450		

Thank You!



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Section: 3CSE1



School of Computer Science &
Engineering

IITM University, Greater Noida, U.P.